



Date: 25 /06 /2026

Lab Practical #01:

Study of basic networking commands and IP configuration.

Practical Assignment #01:

1. Perform and explain various networking commands listed below:
 - i. ipconfig
 - ii. ping
 - iii. getmac
 - iv. systeminfo
 - v. traceroute / tracert
 - vi. netstat
 - vii. nslookup
 - viii. hostname
 - ix. pathping
 - x. arp
 - xi. dig

1. ipconfig

Description:

These commands show your computer's complete network identity — its IP address, subnet mask, default gateway, DNS servers, and MAC address. Think of it as your computer's **network ID card**.

No.	Option	Description
1	ipconfig /displaydns	View cached DNS entries
2	Ipconfig /release	Drop current IP
3	Ipconfig /renew	Request new IP from DHCP
4	Ipconfig / flushdns	Clear DNS cache
5	Ipconfig /all	displays the complete, detailed TCP/IP network configuration for all network adapters on your computer

Implementation:

----- Screenshot Only (Execute command on command prompt / terminal) -----



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```
C:\Users\LENOVO>ipconfig /?

USAGE:
    ipconfig [/allcompartments] [/? | /all |
        /renew [adapter] | /release [adapter] |
        /renew6 [adapter] | /release6 [adapter] |
        /flushdns | /displaydns | /registerdns |
        /showclassid adapter |
        /setclassid adapter [classid] |
        /showclassid6 adapter |
        /setclassid6 adapter [classid] ]

where
    adapter          Connection name
                     (wildcard characters * and ? allowed, see examples)

Options:
    /?              Display this help message
    /all            Display full configuration information.
    /release        Release the IPv4 address for the specified adapter.
    /release6       Release the IPv6 address for the specified adapter.
    /renew          Renew the IPv4 address for the specified adapter.
    /renew6         Renew the IPv6 address for the specified adapter.
    /flushdns       Purges the DNS Resolver cache.
    /registerdns     Refreshes all DHCP leases and re-registers DNS names
    /displaydns     Display the contents of the DNS Resolver Cache.
    /showclassid    Displays all the dhcp class IDs allowed for adapter.
    /setclassid     Modifies the dhcp class id.
    /showclassid6   Displays all the IPv6 DHCP class IDs allowed for adapter.
    /setclassid6    Modifies the IPv6 DHCP class id.
```

```
C:\Users\LENOVO>ipconfig /release

Windows IP Configuration

No operation can be performed on Local Area Connection while it has its media disconnected.
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.

Unknown adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : mshome.net
    Link-local IPv6 Address . . . . . : fe80::301f:5f54:2141:34e3%18
    Default Gateway . . . . . :

C:\Users\LENOVO>
```



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```
C:\Users\LENOVO>ipconfig /renew

Windows IP Configuration

No operation can be performed on Local Area Connection while it has its media disconnected.
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.

Unknown adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : mshome.net
    Link-local IPv6 Address . . . . . : fe80::301f:5f54:2141:34e3%18
    IPv4 Address. . . . . : 10.26.65.157
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 10.26.1.1

C:\Users\LENOVO>
```

```
C:\Users\LENOVO>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

C:\Users\LENOVO>
```



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```
C:\Users\LENOVO>ipconfig /all

Windows IP Configuration

Host Name . . . . . : LAPTOP-DARSHAN
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : mshome.net

Unknown adapter Local Area Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : TAP-Windows Adapter V9
Physical Address. . . . . : 00-FF-14-7D-64-79
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : BA-1E-A4-D7-38-CD
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : mshome.net
Description . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
Physical Address. . . . . : B8-1E-A4-D7-38-CD
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::301f:5f54:2141:34e3%18(Preferred)
IPv4 Address. . . . . : 10.26.65.157(Preferred)
Subnet Mask . . . . . : 255.255.0.0
Lease Obtained. . . . . : 25 June, 2026 10:16:52 AM
Lease Expires . . . . . : 26 June, 2026 10:16:52 AM
Default Gateway . . . . . : 10.26.1.1
DHCP Server . . . . . : 10.26.1.1
DHCPv6 IAID . . . . . : 314056356
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-40-E1-0E-B8-1E-A4-D7-38-CD
DNS Servers . . . . . : fe80::dc79:5388:a8f8:e86c%18
                        8.8.8.8
                        1.1.1.1
NetBIOS over Tcpip. . . . . : Enabled
Connection-specific DNS Suffix Search List :
                        mshome.net

C:\Users\LENOVO>
```



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2. ping

Description:

----- Details About Command -----

ping sends small data packets to a target and waits for a reply. It checks if a device is reachable and measures how long the round-trip takes — this delay is called **latency**, measured in milliseconds (ms).

No.	Option	Description
1	Ping -t youtube.com	Ping the specified host until stopped.
2	Ping -a youtube.com	Resolve addresses to hostnames.
3	Ping -n 15 youtube.com	Number of echo requests to send.
4	Ping -f youtube.com	Sets the Don't Fragment flag; tests whether packets can travel without fragmentation.
5	Ping -l 10 youtube.com	Send the buffer size.

Implementation:

----- Screenshot Only (Execute command on command prompt / terminal) -----

```
C:\Users\LENOVO>color f0
C:\Users\LENOVO>ping /?

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
          [-r count] [-s count] [[-j host-list] | [-k host-list]]
          [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
          [-4] [-6] target_name

Options:
    -t          Ping the specified host until stopped.
                To see statistics and continue - type Control-Break;
                To stop - type Control-C.
    -a          Resolve addresses to hostnames.
    -n count    Number of echo requests to send.
    -l size     Send buffer size.
    -f         Set Don't Fragment flag in packet (IPv4-only).
    -i TTL     Time To Live.
    -v TOS     Type Of Service (IPv4-only). This setting has been deprecated
              and has no effect on the type of service field in the IP
              Header).
    -r count    Record route for count hops (IPv4-only).
    -s count    Timestamp for count hops (IPv4-only).
    -j host-list Loose source route along host-list (IPv4-only).
    -k host-list Strict source route along host-list (IPv4-only).
    -w timeout  Timeout in milliseconds to wait for each reply.
    -R         Use routing header to test reverse route also (IPv6-only).
              Per RFC 5095 the use of this routing header has been
              deprecated. Some systems may drop echo requests if
              this header is used.
    -S srcaddr  Source address to use.
    -c compartment Ping a Hyper-V Network Virtualization provider address.
    -p         Ping using IPv6.
    -4         Force using IPv4.
    -6         Force using IPv6.

C:\Users\LENOVO>
```



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```
C:\Users\LENOVO>ping -t youtube.com

Pinging youtube.com [192.178.193.93] with 32 bytes of data:
Reply from 192.178.193.93: bytes=32 time=38ms TTL=113
Reply from 192.178.193.93: bytes=32 time=25ms TTL=113
Reply from 192.178.193.93: bytes=32 time=26ms TTL=113
Reply from 192.178.193.93: bytes=32 time=25ms TTL=113
Reply from 192.178.193.93: bytes=32 time=26ms TTL=113
Reply from 192.178.193.93: bytes=32 time=27ms TTL=113
Reply from 192.178.193.93: bytes=32 time=30ms TTL=113
Reply from 192.178.193.93: bytes=32 time=26ms TTL=113

Ping statistics for 192.178.193.93:
    Packets: Sent = 8, Received = 8, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 25ms, Maximum = 38ms, Average = 27ms
Control-C
^C
C:\Users\LENOVO>
```

```
C:\Users\LENOVO>ping -f youtube.com

Pinging youtube.com [192.178.193.93] with 32 bytes of data:
Reply from 192.178.193.93: bytes=32 time=47ms TTL=113
Reply from 192.178.193.93: bytes=32 time=28ms TTL=113
Reply from 192.178.193.93: bytes=32 time=27ms TTL=113
Reply from 192.178.193.93: bytes=32 time=26ms TTL=113

Ping statistics for 192.178.193.93:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 26ms, Maximum = 47ms, Average = 32ms

C:\Users\LENOVO>
```

```
C:\Users\LENOVO>ping -a youtube.com

Pinging youtube.com [192.178.193.93] with 32 bytes of data:
Reply from 192.178.193.93: bytes=32 time=37ms TTL=113
Reply from 192.178.193.93: bytes=32 time=26ms TTL=113
Reply from 192.178.193.93: bytes=32 time=26ms TTL=113
Reply from 192.178.193.93: bytes=32 time=27ms TTL=113

Ping statistics for 192.178.193.93:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 26ms, Maximum = 37ms, Average = 29ms

C:\Users\LENOVO>
C:\Users\LENOVO>
```



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```
C:\Users\LENOVO>ping -n 15 youtube.com

Pinging youtube.com [192.178.193.190] with 32 bytes of data:
Reply from 192.178.193.190: bytes=32 time=45ms TTL=112
Reply from 192.178.193.190: bytes=32 time=25ms TTL=112
Reply from 192.178.193.190: bytes=32 time=25ms TTL=112
Reply from 192.178.193.190: bytes=32 time=31ms TTL=112
Reply from 192.178.193.190: bytes=32 time=24ms TTL=112
Reply from 192.178.193.190: bytes=32 time=25ms TTL=112
Reply from 192.178.193.190: bytes=32 time=26ms TTL=112
Reply from 192.178.193.190: bytes=32 time=25ms TTL=112
Reply from 192.178.193.190: bytes=32 time=24ms TTL=112
Reply from 192.178.193.190: bytes=32 time=25ms TTL=112
Reply from 192.178.193.190: bytes=32 time=25ms TTL=112
Reply from 192.178.193.190: bytes=32 time=26ms TTL=112
Reply from 192.178.193.190: bytes=32 time=24ms TTL=112
Reply from 192.178.193.190: bytes=32 time=27ms TTL=112
Reply from 192.178.193.190: bytes=32 time=24ms TTL=112

Ping statistics for 192.178.193.190:
    Packets: Sent = 15, Received = 15, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 24ms, Maximum = 45ms, Average = 26ms

C:\Users\LENOVO>

C:\Users\LENOVO>
```

```
C:\Users\LENOVO>ping -l 10 youtube.com

Pinging youtube.com [192.178.193.190] with 10 bytes of data:
Reply from 192.178.193.190: bytes=10 time=43ms TTL=112
Reply from 192.178.193.190: bytes=10 time=24ms TTL=112
Reply from 192.178.193.190: bytes=10 time=25ms TTL=112
Reply from 192.178.193.190: bytes=10 time=25ms TTL=112

Ping statistics for 192.178.193.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 24ms, Maximum = 43ms, Average = 29ms

C:\Users\LENOVO>
```

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3. getmac

Description:

----- Details About Command -----

getmac shows the MAC (Media Access Control) address of every network adapter on your system. It is the unique hardware identifier of the network card.

No.	Option	Description
1	Getmac / FO csv	Displays MAC addresses in CSV (Comma-Separated Values) format .
2	Getmac /S User /NH /V	Shows MAC addresses from remote system "User" with verbose details and without headers .
3	Getmac/fo list	Displays MAC addresses in list format .
4	Getmac /s systemname /NH /V	Shows MAC addresses of remote computer "systemname" with verbose details and no column headers .
5	Getmac /fo table	Displays MAC addresses in table format .

Implementation:

----- Screenshot Only (Execute command on command prompt / terminal) -----

```
C:\Users\LENOVO>getmac /?
GETMAC [/S system [/U username [/P [password]]]] [/FO format] [/NH] [/V]

Description:
  This tool enables an administrator to display the MAC address
  for network adapters on a system.

Parameter List:
  /S      system          Specifies the remote system to connect to.
  /U      [domain\]user   Specifies the user context under
                          which the command should execute.
  /P      [password]      Specifies the password for the given
                          user context. Prompts for input if omitted.
  /FO     format          Specifies the format in which the output
                          is to be displayed.
                          Valid values: "TABLE", "LIST", "CSV".
  /NH                                           Specifies that the "Column Header" should
                          not be displayed in the output.
                          Valid only for TABLE and CSV formats.
  /V                                           Specifies that verbose output is displayed.
  /?                                           Displays this help message.

Examples:
  GETMAC /?
  GETMAC /FO csv
  GETMAC /S system /NH /V
  GETMAC /S system /U user
  GETMAC /S system /U domain\user /P password /FO list /V
  GETMAC /S system /U domain\user /P password /FO table /NH

C:\Users\LENOVO>|
```



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```
C:\Users\LENOVO>getmac /FO csv
"Physical Address","Transport Name"
"B8-1E-A4-D7-38-CD","\Device\Tcpip_{FDAA482E-48D5-4114-8C87-BA1402A421E9}"
"00-FF-14-7D-64-79","Media disconnected"
```

```
C:\Users\LENOVO>|
```

```
C:\Users\LENOVO>getmac /FO list

Physical Address: B8-1E-A4-D7-38-CD
Transport Name:   \Device\Tcpip_{FDAA482E-48D5-4114-8C87-BA1402A421E9}

Physical Address: 00-FF-14-7D-64-79
Transport Name:   Media disconnected
```

```
C:\Users\LENOVO>|
```

```
C:\Users\LENOVO>getmac /FO table
```

Physical Address	Transport Name
B8-1E-A4-D7-38-CD	\Device\Tcpip_{FDAA482E-48D5-4114-8C87-BA1402A421E9}
00-FF-14-7D-64-79	Media disconnected

```
C:\Users\LENOVO>|
```

```
C:\Users\LENOVO>getmac /S LAPTOP-DARSHAN /NH /V

Wi-Fi           Realtek RTL8852 B8-1E-A4-D7-38-CD   \Device\Tcpip_{FDAA482E-48D5-4114-8C87-BA1402A421E9}
Local Area Conn TAP-Windows Ada 00-FF-14-7D-64-79   Media disconnected
```

```
C:\Users\LENOVO>|
```

```
C:\Users\LENOVO>getmac /S LENOVO /NH /V
ERROR: The RPC server is unavailable.
```



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4. systeminfo

Description:

----- Details About Command -----

systeminfo gives a complete profile of the system — OS version, RAM, CPU, installed patches, and network card details. Useful for diagnosing compatibility or configuration issues.

No.	Option	Description
1	Systeminfo /S systemname	Displays system information of the remote computer " systemname ".
2	Systeminfo /S systemname /U User	Displays remote system information using the specified user account credentials.
3	Systeminfo /S systemname /FO LIST	Shows remote system information in list format .
4	Systeminfo /S systemname /FO CSV /NH	Shows remote system information in CSV format without headers .
5	Systeminfo	Displays detailed information about the local computer's OS, hardware, memory, and network configuration.

Implementation:

----- Screenshot Only (Execute command on command prompt / terminal) -----

```
C:\Users\LENOVO>systeminfo /?
SYSTEMINFO [/S system [/U username [/P [password]]] [/FO format] [/NH]

Description:
  This tool displays operating system configuration information for
  a local or remote machine, including service pack levels.

Parameter List:
  /S      system          Specifies the remote system to connect to.
  /U      [domain\]user   Specifies the user context under which
                          the command should execute.
  /P      [password]      Specifies the password for the given
                          user context. Prompts for input if omitted.
  /FO     format          Specifies the format in which the output
                          is to be displayed.
                          Valid values: "TABLE", "LIST", "CSV".
  /NH                                           Specifies that the "Column Header" should
                          not be displayed in the output.
                          Valid only for "TABLE" and "CSV" formats.
  /?                                           Displays this help message.

Examples:
  SYSTEMINFO
  SYSTEMINFO /?
  SYSTEMINFO /S system
  SYSTEMINFO /S system /U user
  SYSTEMINFO /S system /U domain\user /P password /FO TABLE
  SYSTEMINFO /S system /FO LIST
  SYSTEMINFO /S system /FO CSV /NH

C:\Users\LENOVO>
```



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C:\Users\LENOVO>systeminfo /S LAPTOP-DARSHAN

```
Host Name: LAPTOP-DARSHAN
OS Name: Microsoft Windows 11 Home Single Language
OS Version: 10.0.26200 N/A Build 26200
OS Manufacturer: Microsoft Corporation
OS Configuration: Standalone Workstation
OS Build Type: Multiprocessor Free
Registered Owner: LENOVO
Registered Organization: N/A
Product ID: 00342-42683-03666-AAOEM
Original Install Date: 26-Aug-25, 8:14:00 PM
System Boot Time: 20-Jun-26, 4:14:20 PM
System Manufacturer: LENOVO
System Model: 83ER
System Type: x64-based PC
Processor(s): 1 Processor(s) Installed.
[01]: Intel64 Family 6 Model 154 Stepping 3 GenuineIntel ~2000 Mhz
BIOS Version: LENOVO LTCN31WW, 29-Nov-23
Windows Directory: C:\WINDOWS
System Directory: C:\WINDOWS\system32
Boot Device: \Device\HarddiskVolume1
System Locale: en-us;English (United States)
Input Locale: en-us;English (United States)
Time Zone: (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory: 16,077 MB
Available Physical Memory: 7,464 MB
Virtual Memory: Max Size: 26,317 MB
Virtual Memory: Available: 14,883 MB
Virtual Memory: In Use: 11,434 MB
Page File Location(s): C:\WINDOWS\D
Domain: WORKGROUP
Logon Server: \\LAPTOP-DARSHAN
Hotfix(s): 4 Hotfix(s) Installed.
[01]: KB5087051
[02]: KB5054156
[03]: KB5094126
[04]: KB5094135
Network Card(s): 2 NIC(s) Installed.
[01]: Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
Connection Name: Wi-Fi
DHCP Enabled: Yes
DHCP Server: 10.26.1.1
IP address(es)
[01]: 10.26.65.157
[02]: fe80::301f:5f54:2141:34e3
[02]: TAP-Windows Adapter V9
Connection Name: Local Area Connection
Status: Media disconnected
Virtualization-based security: Status: Running
Required Security Properties:
Available Security Properties:
Base Virtualization Support
Secure Boot
DMA Protection
UEFI Code Readonly
SMM Security Mitigations 1.0
Mode Based Execution Control
APIC Virtualization
Services Configured:
Hypervisor enforced Code Integrity
Services Running:
Hypervisor enforced Code Integrity
App Control for Business policy: Enforced
App Control for Business user mode policy: Off
Security Features Enabled:
Hyper-V Requirements: A hypervisor has been detected. Features required for Hyper-V will not be displayed.
```



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```
C:\Users\LENOVO>systeminfo

Host Name: LAPTOP-DARSHAN
OS Name: Microsoft Windows 11 Home Single Language
OS Version: 10.0.26200 N/A Build 26200
OS Manufacturer: Microsoft Corporation
OS Configuration: Standalone Workstation
OS Build Type: Multiprocessor Free
Registered Owner: LENOVO
Registered Organization: N/A
Product ID: 00342-42683-03666-AAOEM
Original Install Date: 26-Aug-25, 8:14:00 PM
System Boot Time: 20-Jun-26, 4:14:20 PM
System Manufacturer: LENOVO
System Model: 83ER
System Type: x64-based PC
Processor(s): 1 Processor(s) Installed.
[01]: Intel64 Family 6 Model 154 Stepping 3 GenuineIntel ~2000 Mhz
BIOS Version: LENOVO LTCN31WW, 29-Nov-23
Windows Directory: C:\WINDOWS
System Directory: C:\WINDOWS\system32
Boot Device: \Device\HarddiskVolume1
System Locale: en-us;English (United States)
Input Locale: en-us;English (United States)
Time Zone: (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory: 16,077 MB
Available Physical Memory: 7,450 MB
Virtual Memory: Max Size: 26,317 MB
Virtual Memory: Available: 14,849 MB
Virtual Memory: In Use: 11,468 MB
Page File Location(s): C:\WINDOWS\D
Domain: WORKGROUP
Logon Server: \\LAPTOP-DARSHAN
Hotfix(s): 4 Hotfix(s) Installed.
[01]: KB5087051
[02]: KB50854156
[03]: KB5094126
[04]: KB5094135

Network Card(s): 2 NIC(s) Installed.
[01]: Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
Connection Name: Wi-Fi
DHCP Enabled: Yes
DHCP Server: 10.26.1.1
IP address(es)
[01]: 10.26.65.157
[02]: fe80::301f:5f54:2141:34e3
[02]: TAP-Windows Adapter V9
Connection Name: Local Area Connection
Status: Media disconnected

Virtualization-based security: Status: Running
Required Security Properties:
Available Security Properties:
Base Virtualization Support
Secure Boot
DMA Protection
UEFI Code Readonly
SMM Security Mitigations 1.0
Mode Based Execution Control
APIC Virtualization
Services Configured:
Hypervisor enforced Code Integrity
Services Running:
Hypervisor enforced Code Integrity
App Control for Business policy: Enforced
App Control for Business user mode policy: Off
Security Features Enabled:
Hyper-V Requirements: A hypervisor has been detected. Features required for Hyper-V will not be displayed.

C:\Users\LENOVO>
```



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```
C:\Users\LENOVO>systeminfo /S LAPTOP-DARSHAN /U LENOVO
WARNING: User credentials cannot be used for local connections

Host Name:                               LAPTOP-DARSHAN
OS Name:                                 Microsoft Windows 11 Home Single Language
OS Version:                             10.0.26200 N/A Build 26200
OS Manufacturer:                        Microsoft Corporation
OS Configuration:                       Standalone Workstation
OS Build Type:                           Multiprocessor Free
Registered Owner:                        LENOVO
Registered Organization:                  N/A
Product ID:                              00342-42683-03666-AAOEM
Original Install Date:                   26-Aug-25, 8:14:00 PM
System Boot Time:                         20-Jun-26, 4:14:20 PM
System Manufacturer:                     LENOVO
System Model:                             83ER
System Type:                             x64-based PC
Processor(s):                             1 Processor(s) Installed.
                                           [01]: Intel64 Family 6 Model 154 Stepping 3 GenuineIntel ~2000 Mhz
BIOS Version:                             LENOVO LTCN31WW, 29-Nov-23
Windows Directory:                       C:\WINDOWS
System Directory:                         C:\WINDOWS\system32
Boot Device:                             \Device\HarddiskVolume1
System Locale:                             en-us;English (United States)
Input Locale:                             en-us;English (United States)
Time Zone:                               (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory:                    16,077 MB
Available Physical Memory:                 7,466 MB
Virtual Memory: Max Size:                 26,317 MB
Virtual Memory: Available:                14,863 MB
Virtual Memory: In Use:                   11,454 MB
Page File Location(s):                    C:\WINDOWS\D
Domain:                                   WORKGROUP
Logon Server:                             \\LAPTOP-DARSHAN
Hotfix(s):                                4 Hotfix(s) Installed.
                                           [01]: KB5087051
                                           [02]: KB5054156
                                           [03]: KB5094126
                                           [04]: KB5094135
Network Card(s):                          2 NIC(s) Installed.
                                           [01]: Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
                                           Connection Name: Wi-Fi
                                           Status: Media disconnected
                                           [02]: TAP-Windows Adapter V9
                                           Connection Name: Local Area Connection
                                           Status: Media disconnected
Virtualization-based security: Status: Running
Required Security Properties:
Available Security Properties:
    Base Virtualization Support
    Secure Boot
    DMA Protection
    UEFI Code Readonly
    SMM Security Mitigations 1.0
    Mode Based Execution Control
    APIC Virtualization
Services Configured:
    Hypervisor enforced Code Integrity
Services Running:
    Hypervisor enforced Code Integrity
App Control for Business policy: Enforced
App Control for Business user mode policy: Off
Security Features Enabled:
Hyper-V Requirements:                     A hypervisor has been detected. Features required for Hyper-V will not be displayed.

C:\Users\LENOVO>
```



DARSHAN INSTITUTE OF ENGINEERING & TECHNOLOGY

Semester 5th | Practical Assignment | Computer Networks (2301CS501)

Date: 25 /06 /2026

C:\Users\LENOVO>systeminfo /S LAPTOP-DARSHAN /FO list

```
Host Name: LAPTOP-DARSHAN
OS Name: Microsoft Windows 11 Home Single Language
OS Version: 10.0.26200 N/A Build 26200
OS Manufacturer: Microsoft Corporation
OS Configuration: Standalone Workstation
OS Build Type: Multiprocessor Free
Registered Owner: LENOVO
Registered Organization: N/A
Product ID: 00342-42683-03666-AAOEM
Original Install Date: 26-Aug-25, 8:14:00 PM
System Boot Time: 20-Jun-26, 4:14:20 PM
System Manufacturer: LENOVO
System Model: 83ER
System Type: x64-based PC
Processor(s): 1 Processor(s) Installed.
[01]: Intel64 Family 6 Model 154 Stepping 3 GenuineIntel ~2000 Mhz
BIOS Version: LENOVO LTCN31WW, 29-Nov-23
Windows Directory: C:\WINDOWS
System Directory: C:\WINDOWS\system32
Boot Device: \Device\HarddiskVolume1
System Locale: en-us;English (United States)
Input Locale: en-us;English (United States)
Time Zone: (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory: 16,077 MB
Available Physical Memory: 7,437 MB
Virtual Memory: Max Size: 26,317 MB
Virtual Memory: Available: 14,799 MB
Virtual Memory: In Use: 11,518 MB
Page File Location(s): C:\WINDOWS\D
Domain: WORKGROUP
Logon Server: \\LAPTOP-DARSHAN
Hotfix(s): 4 Hotfix(s) Installed.
[01]: KB5087051
[02]: KB5054156
[03]: KB5094126
[04]: KB5094135
Network Card(s): 2 NIC(s) Installed.
[01]: Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
Connection Name: Wi-Fi
DHCP Enabled: Yes
DHCP Server: 10.26.1.1
IP address(es)
[01]: 10.26.65.157
[02]: fe80::301f:5f54:2141:34e3
[02]: TAP-Windows Adapter V9
Connection Name: Local Area Connection
Status: Media disconnected
Virtualization-based security: Status: Running
Required Security Properties:
Available Security Properties:
Base Virtualization Support
Secure Boot
DMA Protection
UEFI Code Readonly
SMM Security Mitigations 1.0
Mode Based Execution Control
APIC Virtualization
Services Configured:
Hypervisor enforced Code Integrity
Services Running:
Hypervisor enforced Code Integrity
App Control for Business policy: Enforced
App Control for Business user mode policy: Off
Security Features Enabled:
Hyper-V Requirements: A hypervisor has been detected. Features required for Hyper-V will not be displayed.
```



Date: 25 /06 /2026

5. traceroute/tracert

Description:

----- Details About Command -----

tracert shows every router (called a **hop**) your data passes through on its way to a destination. It is like getting a receipt of every city your package passed through during delivery.

No.	Option	Description
1	Tracert -h 10 google.com	Traces the route to google.com with a maximum of 10 hops .
2	Tracert -d google.com	Traces the route without resolving IP addresses to host names (faster).
3	Tracert google.com	Displays the path and delay of packets from your computer to google.com .

Implementation:

----- Screenshot Only (Execute command on command prompt / terminal) -----

```
C:\Users\LENOVO>tracert /?

Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
              [-R] [-S srcaddr] [-4] [-6] target_name

Options:
-d          Do not resolve addresses to hostnames.
-h maximum_hops  Maximum number of hops to search for target.
-j host-list  Loose source route along host-list (IPv4-only).
-w timeout    Wait timeout milliseconds for each reply.
-R          Trace round-trip path (IPv6-only).
-S srcaddr    Source address to use (IPv6-only).
-4          Force using IPv4.
-6          Force using IPv6.

C:\Users\LENOVO>
```

```
C:\Users\LENOVO>tracert -h 10 google.com

Tracing route to google.com [192.178.193.113]
over a maximum of 10 hops:

  1  15 ms    2 ms    2 ms  10.26.1.1
  2  17 ms    3 ms    3 ms  static.ill.117.250.171.130.bsnl.co.in [117.250.171.130]
  3  *         *         *    Request timed out.
  4  *         *         *    Request timed out.
  5  *         *         *    Request timed out.
  6  *         *         *    Request timed out.
  7  *         *         *    Request timed out.
  8  17 ms    18 ms    18 ms  72.14.197.4
  9  18 ms    18 ms    18 ms  216.239.47.175
 10  19 ms    19 ms    57 ms  172.253.177.90

Trace complete.

C:\Users\LENOVO>
```



Date: 25 /06 /2026

```
C:\Users\LENOVO>tracert -d google.com

Tracing route to google.com [192.178.193.113]
over a maximum of 30 hops:

  1    8 ms    4 ms    3 ms  10.26.1.1
  2    5 ms    4 ms    5 ms  117.250.171.130
  3    *        *        *    Request timed out.
  4    *        *        *    Request timed out.
  5    *        *        *    Request timed out.
  6    *        *        *    Request timed out.
  7    *        *        *    Request timed out.
  8   23 ms   17 ms   17 ms  72.14.197.4
  9   18 ms   18 ms   29 ms  216.239.47.175
 10   35 ms   18 ms   19 ms  172.253.177.90
 11   18 ms   20 ms   18 ms  142.251.77.255
 12   18 ms   18 ms   18 ms  192.178.45.10
 13    *        *        *    Request timed out.
 14    *        *        *    Request timed out.
 15    *        *        *    Request timed out.
 16    *        *        *    Request timed out.
 17    *        *        *    Request timed out.
 18    *        *        *    Request timed out.
 19   18 ms   17 ms   17 ms  192.178.193.113

Trace complete.

C:\Users\LENOVO>
```

```
C:\Users\LENOVO>tracert google.com

Tracing route to google.com [192.178.193.113]
over a maximum of 30 hops:

  1    4 ms    5 ms    4 ms  10.26.1.1
  2    6 ms    4 ms    4 ms  static.ill.117.250.171.130.bsnl.co.in [117.250.171.130]
  3    *        *        *    Request timed out.
  4    *        *        *    Request timed out.
  5    *        *        *    Request timed out.
  6    *        *        *    Request timed out.
  7    *        *        *    Request timed out.
  8   17 ms   18 ms   18 ms  72.14.197.4
  9   19 ms   19 ms   21 ms  216.239.47.175
 10   17 ms   17 ms   18 ms  172.253.177.90
 11   22 ms   22 ms   19 ms  142.251.77.255
 12   18 ms   25 ms   18 ms  192.178.45.10
 13    *        *        *    Request timed out.
 14    *        *        *    Request timed out.
 15    *        *        *    Request timed out.
 16    *        *        *    Request timed out.
 17    *        *        *    Request timed out.
 18    *        *        *    Request timed out.
 19   24 ms   18 ms   28 ms  ly-in-f113.1e100.net [192.178.193.113]

Trace complete.

C:\Users\LENOVO>
```



Date: 25 /06 /2026

6. netstat

Description:

----- Details About Command -----

netstat shows all active network connections on your machine — which ports are open, which remote servers you are connected to, and which process owns each connection. Very useful for security analysis.

No.	Option	Description
1	Netstat -l	Displays all listening ports on the system
2	Netstat -e	Displays Ethernet statistics , including bytes sent and received
3	Netstat -f	Displays fully qualified domain names (FQDNs) for foreign addresses
4	Netstat -i	Displays network interface statistics
5	Netstat -n	Displays active connections and listening ports using numeric IP addresses and port numbers instead of names.

Implementation:

----- Screenshot Only (Execute command on command prompt / terminal) -----

```
C:\Users\LENOVO>netstat -f

Active Connections

Proto Local Address           Foreign Address         State
TCP   10.26.65.157:51675       20.42.65.91:https       ESTABLISHED
TCP   10.26.65.157:52127       67.205.137.196:https    ESTABLISHED
TCP   10.26.65.157:56012       13.69.239.72:https      ESTABLISHED
TCP   10.26.65.157:56013       24.224.186.35.bc.googleusercontent.com:https ESTABLISHED
TCP   10.26.65.157:56014       52.110.14.142:https     TIME_WAIT
TCP   10.26.65.157:62037       20.192.44.82:https      ESTABLISHED
TCP   10.26.65.157:62226       4.213.25.240:https      ESTABLISHED
TCP   10.26.65.157:63423       20.42.65.91:https       ESTABLISHED
TCP   127.0.0.1:49686          LAPTOP-DARSHAN:49687    ESTABLISHED
TCP   127.0.0.1:49687          LAPTOP-DARSHAN:49686    ESTABLISHED
TCP   127.0.0.1:49688          LAPTOP-DARSHAN:49689    ESTABLISHED
TCP   127.0.0.1:49689          LAPTOP-DARSHAN:49688    ESTABLISHED
TCP   127.0.0.1:49694          LAPTOP-DARSHAN:49695    ESTABLISHED
TCP   127.0.0.1:49695          LAPTOP-DARSHAN:49694    ESTABLISHED

C:\Users\LENOVO>
```



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```
C:\Users\LENOVO>netstat -e
Interface Statistics
```

	Received	Sent
Bytes	11600214	2051880
Unicast packets	7296	7506
Non-unicast packets	49110	924
Discards	0	0
Errors	0	0
Unknown protocols	0	

```
C:\Users\LENOVO>|
```

```
C:\Users\LENOVO>netstat -a
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:135	LAPTOP-DARSHAN:0	LISTENING
TCP	0.0.0.0:445	LAPTOP-DARSHAN:0	LISTENING
TCP	0.0.0.0:5040	LAPTOP-DARSHAN:0	LISTENING
TCP	0.0.0.0:49664	LAPTOP-DARSHAN:0	LISTENING
TCP	0.0.0.0:49665	LAPTOP-DARSHAN:0	LISTENING
TCP	0.0.0.0:49668	LAPTOP-DARSHAN:0	LISTENING
TCP	0.0.0.0:49669	LAPTOP-DARSHAN:0	LISTENING
TCP	0.0.0.0:49670	LAPTOP-DARSHAN:0	LISTENING
TCP	0.0.0.0:49697	LAPTOP-DARSHAN:0	LISTENING
TCP	10.26.65.157:49299	172.64.41.3:https	ESTABLISHED
TCP	10.26.65.157:51138	dns:https	ESTABLISHED

^C

```
C:\Users\LENOVO>|
```

```
C:\Users\LENOVO>netstat -i
```

Active Connections

Proto	Local Address	Foreign Address	State	Time in State (ms)
TCP	10.26.65.157:49299	172.64.41.3:https	TIME_WAIT	35265
TCP	127.0.0.1:49686	LAPTOP-DARSHAN:49687	ESTABLISHED	413534451
TCP	127.0.0.1:49687	LAPTOP-DARSHAN:49686	ESTABLISHED	413534451
TCP	127.0.0.1:49688	LAPTOP-DARSHAN:49689	ESTABLISHED	413533446
TCP	127.0.0.1:49689	LAPTOP-DARSHAN:49688	ESTABLISHED	413533446
TCP	127.0.0.1:49694	LAPTOP-DARSHAN:49695	ESTABLISHED	413530356
TCP	127.0.0.1:49695	LAPTOP-DARSHAN:49694	ESTABLISHED	413530356
TCP	10.26.65.157:51138	dns:https	TIME_WAIT	35271
TCP	10.26.65.157:52044	51.11.192.49:https	TIME_WAIT	35096

^C

```
C:\Users\LENOVO>|
```



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```
C:\Users\LENOVO>netstat -n
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	10.26.65.157:49299	172.64.41.3:443	TIME_WAIT
TCP	10.26.65.157:51138	8.8.4.4:443	TIME_WAIT
TCP	10.26.65.157:52044	51.11.192.49:443	TIME_WAIT
TCP	10.26.65.157:52045	52.230.60.54:443	TIME_WAIT
TCP	10.26.65.157:52046	13.107.6.158:443	ESTABLISHED
TCP	10.26.65.157:52048	52.230.60.54:443	TIME_WAIT
TCP	10.26.65.157:52049	35.186.224.24:443	ESTABLISHED
TCP	10.26.65.157:52050	52.123.253.132:443	ESTABLISHED
TCP	10.26.65.157:52366	172.64.41.3:443	TIME_WAIT
TCP	10.26.65.157:52626	8.8.8.8:443	TIME_WAIT
TCP	10.26.65.157:53281	162.159.61.3:443	TIME_WAIT
TCP	10.26.65.157:54389	8.8.8.8:443	TIME_WAIT
TCP	10.26.65.157:55072	8.8.8.8:443	TIME_WAIT
TCP	10.26.65.157:55363	20.50.80.210:443	ESTABLISHED
TCP	10.26.65.157:55364	142.251.150.119:443	TIME_WAIT
TCP	10.26.65.157:55367	142.250.206.163:443	TIME_WAIT
TCP	10.26.65.157:56491	162.159.61.3:443	TIME_WAIT
TCP	10.26.65.157:57967	162.159.61.3:443	TIME_WAIT
TCP	10.26.65.157:57968	20.184.175.11:443	TIME_WAIT
TCP	10.26.65.157:57969	137.184.103.12:443	ESTABLISHED
TCP	10.26.65.157:58686	162.159.61.3:443	TIME_WAIT
TCP	10.26.65.157:58928	8.8.4.4:443	TIME_WAIT
TCP	10.26.65.157:59128	162.159.61.3:443	TIME_WAIT
TCP	10.26.65.157:60577	8.8.4.4:443	TIME_WAIT
TCP	10.26.65.157:61282	172.64.41.3:443	TIME_WAIT
TCP	10.26.65.157:61283	20.192.44.82:443	TIME_WAIT
TCP	10.26.65.157:61284	20.192.44.82:443	ESTABLISHED
TCP	10.26.65.157:62227	4.213.25.241:443	ESTABLISHED
TCP	10.26.65.157:62705	8.8.8.8:443	TIME_WAIT
TCP	10.26.65.157:62736	8.8.8.8:443	TIME_WAIT
TCP	10.26.65.157:63889	172.64.41.3:443	TIME_WAIT
TCP	10.26.65.157:64130	172.64.41.3:443	TIME_WAIT
TCP	10.26.65.157:64588	8.8.8.8:443	TIME_WAIT
TCP	127.0.0.1:49686	127.0.0.1:49687	ESTABLISHED
TCP	127.0.0.1:49687	127.0.0.1:49686	ESTABLISHED
TCP	127.0.0.1:49688	127.0.0.1:49689	ESTABLISHED
TCP	127.0.0.1:49689	127.0.0.1:49688	ESTABLISHED
TCP	127.0.0.1:49694	127.0.0.1:49695	ESTABLISHED
TCP	127.0.0.1:49695	127.0.0.1:49694	ESTABLISHED

```
C:\Users\LENOVO>
```



Date: 25 /06 /2026

7. nslookup

Description:

----- Details About Command -----

nslookup queries DNS servers to convert domain names into IP addresses. Think of DNS like a phone book — you give a name, it gives back the number (IP address).

No.	Option	Description
1	Nslookup google.com	Displays the IP address associated with google.com .
2	Nslookup	Opens the interactive DNS query mode for manual lookups.
3	Nslookup -type=mx google.com	Displays the Mail Exchange (MX) records of google.com .
4	Nslookup -type=ns google.com	Displays the Name Server (NS) records of google.com
5	Nslookup -type=txt google.com	Displays the TXT records of google.com

Implementation:

----- Screenshot Only (Execute command on command prompt / terminal) -----

```
C:\Users\LENOVO>nslookup -type=ns google.com
Server:  DESKTOP-VAATTS9
Address:  fe80::dc79:5388:a8f8:e86c

Non-authoritative answer:
google.com      nameserver = ns3.google.com
google.com      nameserver = ns1.google.com
google.com      nameserver = ns2.google.com
google.com      nameserver = ns4.google.com

C:\Users\LENOVO>|
```

```
C:\Users\LENOVO>nslookup -type=mx google.com
Server:  DESKTOP-VAATTS9
Address:  fe80::dc79:5388:a8f8:e86c

Non-authoritative answer:
google.com      MX preference = 10, mail exchanger = smtp.google.com

C:\Users\LENOVO>|
```



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```
C:\Users\LENOVO>nslookup
Default Server: UnKnown
Address: 10.25.1.1

> set type=A
> google.com
Server: UnKnown
Address: 10.25.1.1

Non-authoritative answer:
Name: google.com
Addresses: 142.250.146.102
          142.250.146.113
          142.250.146.139
          142.250.146.138
          142.250.146.100
          142.250.146.101
```

```
C:\Users\LENOVO>nslookup -type=txt google.com
Server: UnKnown
Address: 10.37.195.145

Non-authoritative answer:
google.com text =
"MS=E4A68B9AB2BB9670BCE15412F62916164C0B20BB"
google.com text =
"facebook-domain-verification=22rm551cu4k0ab0bxsw536tlds4h95"
google.com text =
"docusign=1b0a6754-49b1-4db5-8540-d2c12664b289"
google.com text =
"cisco-ci-domain-verification=47c38bc8c4b74b7233e9053220c1bbe76bcc1cd33c7acf7acd36cd6a5332004b"
google.com text =
"google-site-verification=wD8N7i1JTNTkezJ49swvWw48f8_9xveREV4oB-0Hf5o"
google.com text =
"google-site-verification=4ibFUGB-wXLQ_S7vsXVomSTVamu0XBivAZpR5IZ87D0"
google.com text =
"onetrust-domain-verification=6d685f1d41a94696ad7ef771f68993e0"
google.com text =
"Z29vZ2xl"
google.com text =
"v=spf1 include:_spf.google.com ~all"
google.com text =
"onetrust-domain-verification=0d477fe608074e6f9c12bca7826035cc"
google.com text =
"docusign=05958488-4752-4ef2-95eb-aa7ba8a3bd0e"
google.com text =
"globalsign-smime-dv=CDYX+XFHuw2wml6/Gb8+59BsH31KzUr6c1l2BPvqKX8="
google.com text =
"apple-domain-verification=30afIBcvSuDV2PLX"
google.com text =
"google-site-verification=TV9-DBe4R80X4v0M4U_bd_J9cp0JM0nikft0jAgjmsQ"

C:\Users\LENOVO>
```

```
C:\Users\LENOVO>nslookup google.com
Server: UnKnown
Address: 10.37.195.145

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4009:80e::200e
          192.178.193.113
          192.178.193.102
          192.178.193.100
          192.178.193.138
          192.178.193.101
          192.178.193.139

C:\Users\LENOVO>
```



Date: 25 /06 /2026

8. hostname

Description:

----- Details About Command -----

hostname prints the name given to your computer on the network. Instead of remembering an IP address, other computers on the same network can use this name to identify your machine.

No.	Option	Description
1	Hostname /?	Prints the name of the current host

Implementation:

----- Screenshot Only (Execute command on command prompt / terminal) -----

```
C:\Users\LENOVO>hostname  
LAPTOP-DARSHAN  
  
C:\Users\LENOVO>|
```

9. pathping

Description:

----- Details About Command -----

pathping combines ping and tracert. It first maps all hops like tracert, then sends 100 packets to each hop and calculates the exact packet loss at every point in the path. It is the most detailed built-in network diagnostic tool on Windows.

No.	Option	Description
1	Pathping google.com	Traces the route to google.com and analyzes packet loss at each hop
2	Pathping -g google.com	Uses a loose source route while tracing the path (IPv4 only; rarely used).
3	Pathping -h 2	Limits the trace to a maximum of 2 hops .
4	Pathping -4 google.com	Forces IPv4 when tracing and analyzing the route.
5	Pathping github.com	Traces the route to github.com and reports network latency and packet loss.



Date: 25 /06 /2026

Implementation:

----- Screenshot Only (Execute command on command prompt / terminal) -----

```
C:\Users\LENOVO>pathping google.com

Tracing route to google.com [2404:6800:4009:80e::200e]
over a maximum of 30 hops:
  0  LAPTOP-DARSHAN [2409:40c1:31d9:b138:d8de:ed38:8264:868]
  1  2409:40c1:31d9:b138::a1
  2  * * *
Computing statistics for 25 seconds...
^C
C:\Users\LENOVO>
```

```
C:\Users\LENOVO>pathping /?

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
               [-p period] [-q num_queries] [-w timeout]
               [-4] [-6] target_name
```

```
Options:
  -g host-list      Loose source route along host-list.
  -h maximum_hops   Maximum number of hops to search for target.
  -i address        Use the specified source address.
  -n               Do not resolve addresses to hostnames.
  -p period         Wait period milliseconds between pings.
  -q num_queries    Number of queries per hop.
  -w timeout        Wait timeout milliseconds for each reply.
  -4               Force using IPv4.
  -6               Force using IPv6.
```

```
C:\Users\LENOVO>
```

```
C:\Users\LENOVO>pathping github.com

Tracing route to github.com [64:ff9b::14cf:4952]
over a maximum of 30 hops:
  0  LAPTOP-DARSHAN [2409:40c1:31d9:b138:d8de:ed38:8264:868]
  1  2409:40c1:31d9:b138::a1
  2  * * *
Computing statistics for 25 seconds...
```



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```
C:\Users\LENOVO>pathping -g google.com

Tracing route to google.com [142.251.222.78]
over a maximum of 30 hops:
  0  LAPTOP-DARSHAN [10.37.195.86]
  1  * * *
Computing statistics for 0 seconds...
Hop  RTT      Source to Here  This Node/Link  Address
  0                               LAPTOP-DARSHAN [10.37.195.86]

Trace complete.

C:\Users\LENOVO>
```

```
C:\Users\LENOVO>pathping -h 2 google.com

Tracing route to google.com [2404:6800:4002:828::200e]
over a maximum of 2 hops:
  0  LAPTOP-DARSHAN [2409:40c1:31d9:b138:d8de:ed38:8264:868]
  1  2409:40c1:31d9:b138::a1
  2  * * *
Computing statistics for 25 seconds...
```

```
C:\Users\LENOVO>pathping -4 google.com

Tracing route to google.com [142.250.193.46]
over a maximum of 30 hops:
  0  LAPTOP-DARSHAN [10.37.195.86]
  1  10.37.195.145
  2  192.0.0.1
  3  * * *
Computing statistics for 50 seconds...
```



Date: 25 /06 /2026

10.arp

Description:

----- Details About Command -----

ARP (Address Resolution Protocol) translates IP addresses into MAC addresses within a local network. Before your computer can send data to 192.168.1.1 (your router), it needs to know its MAC address — ARP handles this automatically.

No.	Option	Description
1	Arp -a	Displays the ARP (Address Resolution Protocol) table , showing the mapping between IP addresses and MAC addresses on the local network.

Implementation:

----- Screenshot Only (Execute command on command prompt / terminal) -----

```
C:\Users\LENOVO>arp -a

Interface: 10.37.195.86 --- 0x12
Internet Address      Physical Address      Type
10.37.195.145         5a-16-64-5d-2d-11    dynamic
10.37.195.255         ff-ff-ff-ff-ff-ff    static
224.0.0.2             01-00-5e-00-00-02    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static

C:\Users\LENOVO>|
```



Date: 25 /06 /2026

```
C:\Users\LENOVO>arp /?
```

Displays and modifies the IP-to-Physical address translation tables used by address resolution protocol (ARP).

```
ARP -s inet_addr eth_addr [if_addr]
```

```
ARP -d inet_addr [if_addr]
```

```
ARP -a [inet_addr] [-N if_addr] [-v]
```

-a Displays current ARP entries by interrogating the current protocol data. If inet_addr is specified, the IP and Physical addresses for only the specified computer are displayed. If more than one network interface uses ARP, entries for each ARP table are displayed.

-g Same as -a.

-v Displays current ARP entries in verbose mode. All invalid entries and entries on the loop-back interface will be shown. Specifies an internet address.

inet_addr Specifies an internet address.

-N if_addr Displays the ARP entries for the network interface specified by if_addr.

-d Deletes the host specified by inet_addr. inet_addr may be

wildcarded with * to delete all hosts.

-s Adds the host and associates the Internet address inet_addr with the Physical address eth_addr. The Physical address is

given as 6 hexadecimal bytes separated by hyphens. The entry is permanent.

eth_addr Specifies a physical address.

if_addr If present, this specifies the Internet address of the interface whose address translation table should be modified.

If not present, the first applicable interface will be used.

Example:

```
> arp -s 157.55.85.212 00-aa-00-62-c6-09 .... Adds a static entry.
```

```
> arp -a .... Displays the arp table.
```

```
C:\Users\LENOVO>
```